UNIVERSITÉ PARIS 1
PANTHÉON SORBONNE

Econometric Report

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1. Introduction

1.1. Research Question

Market capitalization reflects the value that the market assigns to a company and serves as a key indicator for guiding capital allocation decisions. Understanding its determinants is essential for improving the decision-making process of institutional investors. Indeed, these investors must select the most promising securities, allocate capital efficiently across different companies and sectors, and manage the overall risk of their portfolios.

Identifying the organizational and financial factors that influence a company's valuation not only allows investors to anticipate future performance, but also to adjust portfolio weights according to risk profiles and make informed decisions regarding diversification and hedging strategies.

Analyzing these determinants in the period preceding a major financial crisis,here, before 2008, provides a benchmark for the structural factors driving valuation under "normal" market conditions. It also offers a reference point for assessing corporate vulnerability during periods of economic stress or financial turmoil.

This study therefore aims to answer the following question : What are the main organizational and financial determinants of a company's market capitalization ?

1.2. Theoretical Expectations

According to classical discounted cash-flow models, market capitalization reflects the present value of expected future profits, adjusted by the cost of capital. In practice, the higher a company's expected profits and the lower its financing costs or risks, the higher its market value. This approach establishes a direct link between valuation and actual or anticipated financial performance, such as profitability or investment intensity.

Corporate governance may influence firm valuation by limiting conflicts of interest and improving capital allocation (Jensen & Meckling). Board size and diversity are key: overly large boards can reduce efficiency, while diversity can enhance decision quality. Visible governance features can also serve as credibility signals, supporting market value.

Human capital may be another key determinant. Companies with a more skilled workforce or recognized management are generally more innovative and productive, which translates into higher future profits and increased valuation.

The firm life cycle introduces a nuance: young companies may benefit from a growth premium linked to their expansion prospects, whereas more mature firms, even if stable, may be less dynamic. Firm age can therefore have ambiguous effects, depending on whether it reflects experience and credibility or, on the contrary, a slowdown and saturation of growth opportunities.

Finally, financial structure could strongly affect market value. According to capital structure theory (Modigliani–Miller), leverage can provide benefits, such as tax advantages and stronger managerial discipline, but excessive debt increases bankruptcy risk and financial distress costs. It can also create agency problems, underinvestment, or asset substitution. Short-term debt illustrates this dual effect: it may improve discipline but also raises liquidity and refinancing risks, potentially lowering valuation if perceived as threatening by the market.

1.3. Impact Evaluation Methodology

To study the determinants of market capitalization, we propose using a firm-level fixed effects panel data model. This approach allows us to control for unobserved heterogeneity that is constant over time, which corresponds to many structural characteristics of firms (such as managerial quality, corporate culture, or strategy) that are not directly measurable but significantly influence market capitalization. By absorbing these factors into the individual effect, the fixed effects model reduces omitted variable bias related to these time-invariant characteristics.

Using panel data also makes it possible to isolate the within-firm effect of evolving variables such as investment, profitability, or board composition. The estimation then focuses on how changes in these characteristics within a given firm affect its market value, while implicitly controlling for all unobserved time-invariant factors. Adding time fixed effects further neutralizes shocks common to all firms, improving the identification of the determinants of market capitalization. The small number of observed years ($T = 2$) reinforces the relevance of this approach, which efficiently captures invariant heterogeneity while exploiting within-firm variation.

A simple OLS model is not suitable for this analysis for several reasons. First, it suffers from omitted variable bias related to unobserved firm characteristics that OLS cannot correct. Second, many financial variables are endogenous: performance can affect market capitalization, while capitalization can also influence a firm's financial conditions, violating OLS's strict exogeneity assumption. Moreover, OLS does not account for the hierarchical structure of the data (firms over years) and the correlation of errors within firms, nor for the simultaneous correlation among explanatory variables such as investment, profitability, or leverage. Finally, a simple OLS model cannot properly incorporate macroeconomic or sectoral shocks that affect all firms simultaneously. A fixed effects panel model with time fixed effects overcomes these limitations by exploiting within-firm variation and neutralizing both constant factors and common shocks, providing a more robust identification of the determinants of market capitalization.

1.4. Key Findings

Our analysis shows that once firm-specific unobserved heterogeneity is controlled for through fixed effects, only within-firm changes in profitability and investment significantly affect market capitalization. Structural characteristics such as size, governance, or human capital appear influential in pooled OLS models but lose significance in fixed-effects estimations, indicating that their apparent impact reflects between-firm differences rather than true dynamic effects. The heterogeneous-effects analysis further reveals that the market does not value profit uniformly: profit increases have no significant impact for young firms, are positively valued for mature and old firms, and even show negative effects in certain sectors. This confirms that investor interpretation of financial performance depends on firm maturity and industry context.

2. Empirical Analysis

2.1. Dataset Overview

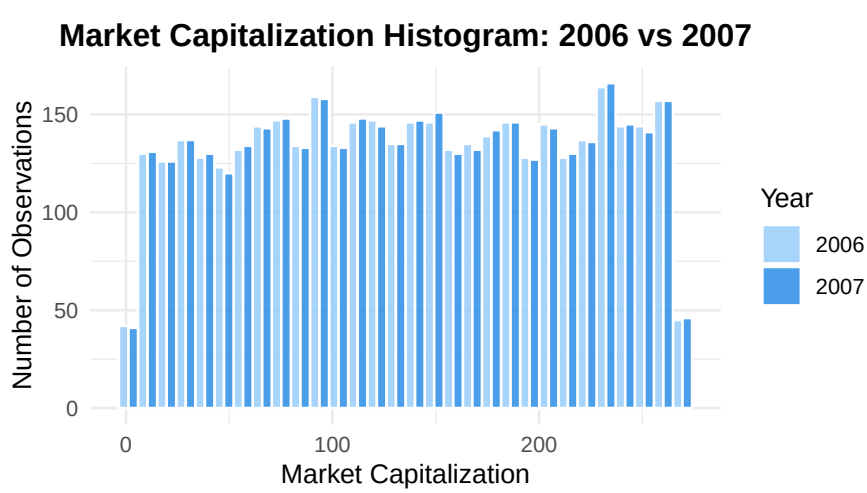
The dataset used in this analysis covers firm-level observations for the years 2006 and 2007, including a total of 8,000 firm-year observations corresponding to 4,000 unique firms followed over two consecutive years. Each firm is observed in both years, which allows us to take advantage of both the cross-sectional variation across firms and the temporal variation within firms. This repeated observation of the same firms over time characterizes the dataset as panel data, making it well-suited for econometric methods that account for unobserved, time-invariant firm characteristics.

The dataset contains important variables such as firm age, sector, number of employees, profit, investment, market capitalization, short-term debt, board size, the number of women on the board and the number of employees with higher education degrees (the variables profit, investment, short-term debt, and market capitalization are expressed in millions of euros). Table 1 presents the main descriptive statistics for these variables. There is sufficient variability both across firms and over time, which is crucial for applying panel data techniques effectively.

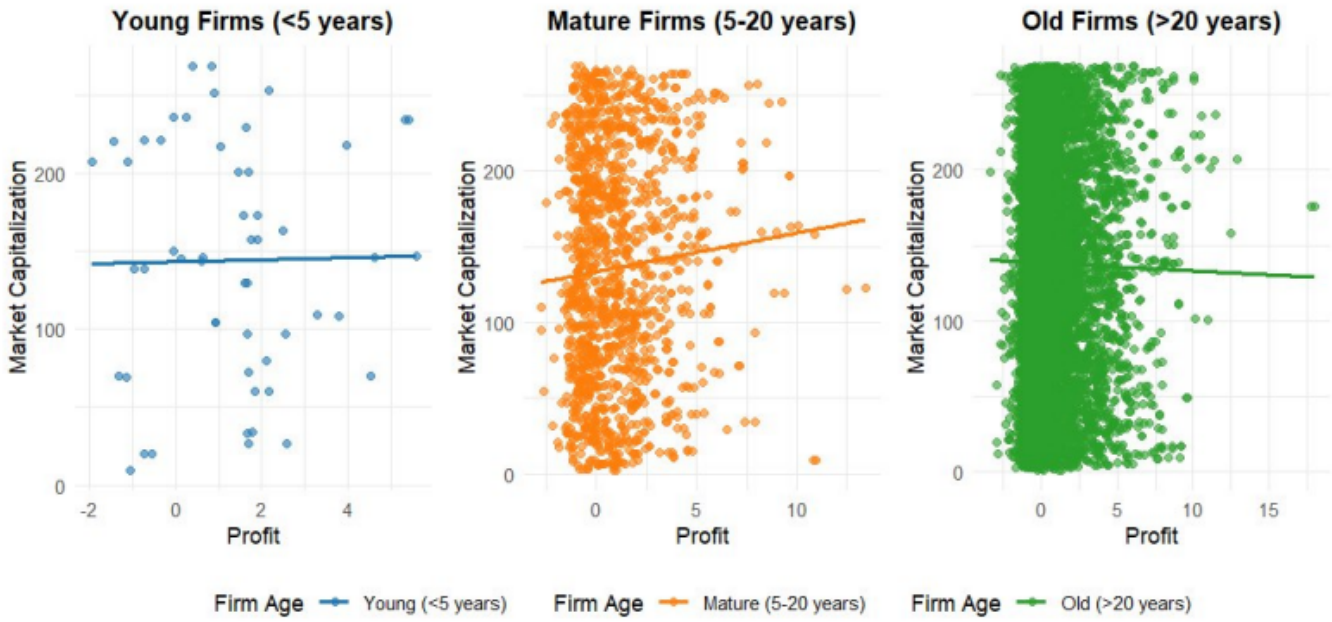
Table 1: Descriptive Statistics

Variable	Mean	Median	Min	Max	Std. Dev.	Skewness	Kurtosis
sector	5.587	6.000	1.000	10.000	2.860	-0.048	4.780
emp	18263.238	18249.000	104.000	35868.000	10301.659	-0.020	4.815
age	29.877	30.000	2.000	67.000	10.125	0.063	5.909
profit	0.939	0.430	-3.380	17.971	2.036	1.528	9.798
inv	22.293	22.490	0.000	62.000	10.606	-0.023	6.041
mktcap	137.697	137.876	1.231	269.451	76.242	-0.020	4.816
shortdebt	21.997	21.995	15.823	26.737	1.093	-0.084	6.533
nboard	99.976	100.000	82.000	118.000	5.058	0.006	5.983
wboard	12.189	7.000	0.000	98.000	13.577	2.313	12.280
skemp	1370.691	1042.500	0.000	5719.000	1180.694	0.959	6.189

Looking more closely at market capitalization, its distribution appears broadly similar between 2006 and 2007, but some differences emerge in the tails and peak frequencies. This suggests that while many firms experience relatively stable market valuations, a substantial number undergo significant changes over time. Such temporal variation underlines the importance of controlling for unobserved, firm-specific factors, which would not be captured in a simple cross-sectional OLS model.



When examining the relationship between profit and market capitalization, it becomes clear that this relationship is not uniform across all firms. Grouping firms by age (young (<5 years), medium (5–20 years), and old (>20 years)) reveals that while profit generally has a positive influence on market capitalization, in some subgroups this effect is weak and might be negative. This observation suggests that additional firm characteristics, such as age, investment, and board composition may play a role in determining market valuation and therefore need to be included as control variables in the econometric models.



Overall, the combination of repeated observations and heterogeneous relationships between profit and market capitalization across firm subgroups motivates the use of a panel data approach with fixed effects. This approach allows us to account for unobserved, time-invariant firm characteristics that could bias simple OLS estimates, providing a more reliable understanding of how profit and other factors influence market capitalization.

2.2. Econometric Specification

Before detailing our econometric strategy, we conducted a series of preliminary tests to determine the most appropriate econometric specification.

The Breusch-Pagan test was first used as a preliminary check for heteroskedasticity, i.e., to assess whether the variance of the residuals is constant across units in a pooled OLS setting. The results suggest that the residual variance is not constant across firms, indicating the presence of heteroskedasticity.

The Hausman test was then applied to determine whether individual effects are correlated with the explanatory variables. The null hypothesis (H0) states that the random-effects model is consistent. The test results, which compare the consistency of the fixed-effects (FE) and random-effects (RE) estimators, led to the rejection of the null hypothesis. This indicates that the random-effects estimator would be inconsistent, confirming the need to adopt a fixed-effects model. This model accounts for unobservable variables that are constant over time and correlated with the explanatory factors, while exploiting intra-individual variation over time to estimate unbiased coefficients. Detailed results of these preliminary tests are presented in the appendix.

Moreover, panel data may exhibit intra-unit correlation of errors, which can bias standard statistical tests. To correct for both heteroskedasticity and intra-unit correlation, we employ robust standard errors clustered by firm. This approach adjusts the standard errors of the estimators so that they remain valid even in the presence of heteroskedasticity or intra-unit correlation, ensuring unbiased coefficient estimates and reliable statistical inference.

Although the preliminary tests revealed the limitations of pooled OLS estimation, we retain this first step because it provides a baseline for comparing different specifications. The main econometric interest lies in observing how the coefficients evolve when individual and temporal fixed effects are introduced, which allows us to assess the robustness of the results and identify the most reliable effects of the explanatory variables. Moreover, since the research question does not assume a direct causal relationship upfront, estimating both pooled OLS and fixed-effects models helps to distinguish correlations driven by structural differences between firms from the causal effects that can be identified through within-firm variation over time.

Progressive pooled OLS: we start by estimating OLS regressions by gradually introducing financial and organizational explanatory variables. This step allows us to observe the evolution of the coefficients and identify the most robust

determinants of the outcome studied.

Individual fixed effects (unit fixed effects / within model): Next, we introduce unit-specific fixed effects to control for structural differences specific to each entity (size, sector, local policy, etc.) and unobservable variables constant over time but correlated with the explanatory variables. Econometrically, this involves transforming the data according to the within estimator, which removes the individual constants without explicitly estimating them, and focuses on intra-individual variation over time. This approach provides unbiased coefficients for the time-varying variables.

Combined model with temporal fixed effects: Finally, we add temporal fixed effects to capture variations common to all units over time (macroeconomic changes, national policies, global trends). This is done by including binary variables for each year, which absorb the influence of temporal factors identical across all units. This final specification allows us to isolate the effect of the explanatory variables while accounting for both structural differences between units and overall temporal variations, ensuring more rigorous and comparable estimates across models.

The final econometric specification can be expressed as:

$$Y_{it} = \alpha_i + \lambda_t + \beta X_{it} + \varepsilon_{it}$$

Y_{it} = log_mktcap$_{it}$, market capitalization of firm i in year t

α_i = Firm fixed effects (capture all time-invariant unobserved characteristics of each firm)

λ_t = Time fixed effects (capture macroeconomic shocks common to all firms)

ε_{it} = Error term, with robust correction for heteroskedasticity and autocorrelation

X_{it} = Financial and organizational explanatory variables: profit$_{it}$, log_inv$_{it}$, log_shortdebt$_{it}$, log_emp$_{it}$, log_board$_{it}$, wboard_ratio$_{it}$, highdegree_emp_ratio$_{it}$, age$_{it}$, age_squared$_{it}$, factor(sector)$_i$

Econometric notes:

- Some variables are transformed into logarithms (log_mktcap, log_emp, log_inv, log_shortdebt, log_board) to allow interpretation in terms of elasticities: a coefficient β_k on a log-transformed variable indicates the percentage effect of a 1% change in that variable on market capitalization. The variable profit$_{it}$ is left in levels because some firms have negative profits.
- We include age$^2_{it}$ to test the non-linear effect of firm age on market capitalization. In fixed effects, we do not include age$_{it}$ directly, as each firm mechanically ages by one year relative to the previous year, which does not provide relevant information; however, age$^2_{it}$ captures the increasing or decreasing effect of accumulated experience.
- The ratios of women on the board wboard_ratio$_{it}$ and of employees with a high level of education highdegree_emp_ratio$_{it}$ are included to analyze relative characteristics of the firm's structure, independently of firm size.
- Variables that are time-invariant, such as log_emp and factor(sector), are automatically excluded from the estimation in a firm fixed effects model, because fixed effects absorb all variation that is constant over time for each firm.

This progressive strategy allows both to compare the results across different specifications, to test the robustness of the coefficients, and to isolate the effect of the explanatory variables while correcting for biases related to individual and time effects.

2.3. Main Results

In the following analysis, the coefficients are interpreted as average effects, ceteris paribus. The regression table is provided in the appendix

Estimates from pooled OLS models provide an initial insight into the determinants of market capitalization but quickly show their limitations when structural and organizational variables are omitted. In Pooled Model 1, which includes only financial variables (profit, investment, and short-term debt), none of the coefficients are significant. The lack of statistical effects and the high constant suggest that variation between firms largely dominates capitalization dynamics, and that short-term financial variables alone are insufficient to explain differences in valuation. This partial specification highlights the importance of including structural characteristics to avoid omitting relevant variables.

The addition of organizational and size-related variables in Pooled Model 2 drastically changes the results. The number of employees becomes by far the main determinant: a 1% increase in a company's number of employees is associated with an approximately 0.91% increase in its market capitalization, on average, ceteris paribus. The effect is not only substantial but also consistent with theoretical models linking size, productive capacity, and financial valuation. Investment also becomes highly significant: a 1% increase in a company's investment corresponds to an approximately 0.02% increase in its market capitalization, all else being equal. In contrast, governance variables (board size and proportion of women) remain insignificant, suggesting that while they may be relevant for organizational performance, these characteristics are not valued by the market in this broad

cross-sectional configuration. Profit becomes significant at the 1% level: A company that makes an additional one million euros in profit sees its market capitalization increase by 0.171%.

The introduction of human capital in Pooled Model 3 adds an additional nuance. The ratio of highly skilled employees slightly increases the explanatory power of the model, but its effect appears negative and marginally significant: a one-percentage-point increase in the ratio of highly skilled employees to the total number of employees in a firm is associated with a slight decrease of 0.035% in its market capitalization, all else being equal, suggesting that a more skilled workforce does not systematically lead to an immediate valuation increase. This inverse relationship, counter-intuitive with respect to human capital theory, may reflect several mechanisms: high labor costs not immediately offset by productivity gains, skill-intensive sectors that are weakly valued, or cycles of investment in human capital whose effects on performance are not instantaneous. This result therefore indicates that a more skilled workforce is not systematically correlated with immediate market valuation, and its impact may be delayed or non-linear.

The inclusion of sector and age effects (Model 4) further refines the interpretation of the determinants of market capitalization. A firm with one additional year of age is associated with an increase in its market capitalization of 0.061% (semi-elasticity), on average, ceteris paribus. However, the low significance of the age-squared variable indicates a slight concave relationship between a firm's age and its market capitalization, suggesting that the marginal effect of age on market capitalization decreases, with a turning point of the estimated polynomial curve around 31 years (the calculation is detailed in the appendix). In other words, market capitalization tends to increase with age up to approximately 31 years, then slightly declines, which is consistent with a life-cycle logic: rapid initial growth followed by maturity and slowdown.

At the sectoral level, certain sectors stand out. Sectors 7 and 8 show a significant effect: firms in sector 8 have a market capitalization approximately 4.45% higher than those in sector 1 (the reference category), all else being equal. In contrast, firms in sector 7 have a market capitalization about 0.73% lower than firms in sector 1, all else being equal.

The effects of the significant variables in the previous models remain largely stable in Model 4, although investment slightly increased, which represents an attempt to improve the robustness of the estimates at the inter-firm level by accounting for various dimensions that may influence capitalization.

Switching to fixed-effects models allows isolating within-firm variation and identifying the determinants that are truly dynamic for market capitalization. The Within model reveals that almost all the effects observed in pooled models stem from structural differences between firms. Only profit and investment remain significant.

At the within-firm level, a 1% increase in a firm's investment is associated with a 0.012% increase in its market capitalization, whereas it was associated with a 0.036% higher market capitalization in Pooled Model 4, on average, ceteris paribus. Thus, approximately two-thirds of the investment effect observed in pooled models can be explained by permanent differences between firms (between effects), and only one-third reflects within-firm dynamics. This substantial reduction suggests that firms with higher market capitalization systematically invest more, but annual variations in investment within a single firm have a much smaller effect on valuation.

In contrast, the effect of profit remained relatively stable: at the within-firm level, a firm that earns an additional one million euros in profit sees its market capitalization increase by 0.188%, compared to a 0.19% increase in Pooled Model 4, on average, ceteris paribus. Profit is thus the most dynamic and "powerful" determinant within a single firm which indicates that internal profit growth indeed translates into higher valuation, in line with discounted cash-flow models.

Conversely, all other determinants lose their significant effect, confirming that their apparent effect in the pooled models came from differences between firms rather than from variations within-firms. This does not imply that they do not influence market capitalization; rather, it indicates that their impact is primarily related to structural and permanent characteristics of each firm, rather than to annual or cyclical variations within a given firm. In other words, these determinants explain the differences in market capitalization between firms, but not necessarily within-firm fluctuations, from year to year.

The model combining fixed effects at the firm-level and in time constitutes the most robust specification, as it simultaneously neutralizes unobserved effects of the firm-specific and macroeconomic shocks common to all years. The results confirm the stability of the two key determinants: investment and profit remain significant and maintain the same coefficients as in the pure FE model . The absence of a significant temporal effect (year 2007) indicates that this year does not present a specific macroeconomic shock in the sample. Organizational, social, and structural variables remain non-significant, confirming their limited role in intra-firm valuation dynamics. This model therefore shows that markets do not react to annual variations in governance or human capital, but primarily value changes in profit and investment.

2.4. Discussion about findings

The general conclusion drawn from our econometric analysis is that market capitalization determinants exhibit significant heterogeneity, operating differently in the cross-sectional dimension (between firms) versus the temporal dynamics within the firm. Pooled OLS estimates were misleading as they primarily reflected permanent structural differences between firms, failing to distinguish them from dynamic effects. This distinction underscores the importance of using an appropriate econometric specification, as a simple OLS model would lead to biased conclusions by conflating structural effects with dynamic effects.

The use of Fixed-Effects models allowed us to eliminate time-invariant unobserved heterogeneity, confirming that within a firm, it is primarily variations in investment and profit that drive market value. This effect highlights the mechanisms at stake in valuation: the market interprets a change in investment (inv) as a strong signal of future growth opportunities, validating theoretical models like Tobin's Q. Conversely, organizational characteristics like governance (wboard_ratio) and human capital largely lose their significance in the FE model, indicating that they primarily influence relative, long-term positions between firms in the cross-section.

To further explore whether these mechanisms operate uniformly across firms, we investigate potential heterogeneous effects by interacting profit with key structural characteristics such as firm age and sector.

Table 3 highlights that the effect of profit on market capitalization varies depending on firm maturity. The sample was divided into three subgroups: young firms ($<$ 5 years), mature firms (5–20 years), and old firms ($>$ 20 years). For young firms, profit variation is not statistically significant, indicating that a €1 million increase in profit does not translate into any observable change in market capitalization, all else being equal. In contrast, mature firms experiencing a €1 million increase in profit see their market capitalization rise by approximately 0.14–0.15% relative to young firms, all else equal. For old firms, the effect is even stronger: an additional €1 million in profit is associated with a roughly 0.18% increase in market capitalization compared to young firms. These results show that the market places a higher value on the profit of firms with a longer and more stable track record. Interestingly, the relationships observed between profit and market capitalization by firm age in the "dataset overview" section do not appear here. This indicates that the initial descriptive correlations did not reflect the true underlying economic relationships, and it confirms the importance of using a Fixed Effects specification to isolate within-firm dynamics from structural differences across firms.

Regarding sectoral heterogeneity, only the interaction between profit and sector 9 is clearly significant and economically meaningful. Firms in this sector exhibit a negative market sensitivity to profit: a €1 million increase in profit is associated with a decrease of about 0.16% in market capitalization compared to firms in sector 1 (the reference category), all else equal. This effect suggests that, in this specific sector, profit is interpreted differently by investors—possibly as a less informative signal of future performance, or as a consequence of sector-specific dynamics where other indicators (such as innovation, market share, or revenue volatility) play a more central role in valuation.

The inclusion of interaction terms serves as a robustness check: it confirms that the dynamic effect of profit remains significant for mature and old firms even after controlling for sectoral and temporal effects. The lack of significance for the 2007 year dummy indicates that no particular macroeconomic shock biases the results. Robustness is thus supported by the stability of the main coefficients (profit and investment) and by the consistency of the profit $\times$ age interactions, showing that the relationship between profit and market capitalization is structurally heterogeneous and cannot be captured by a simple OLS model.

However, several important limitations of our approach should be noted. First, some unobserved variables may vary over time and be correlated with both market capitalization and certain explanatory variables, generating a bias that standard fixed-effects models cannot eliminate. For example, changes in management strategy or sector-specific shocks may simultaneously affect investment and market valuation without being captured by our control variables.

Next, fixed-effects models eliminate time-invariant variables. In theory, their effect can be partially recovered through interactions with time dummies, but with only two periods (T=2), this identification is very limited, capturing only a single shock between the two years.

Moreover, fixed-effects estimation is sensitive to measurement error, which can lead to attenuation bias: the estimated coefficients may underestimate the true effect of the explanatory variables. Therefore, when a coefficient is significant in the FE model, the actual effect can reasonably be considered at least of this magnitude, if not larger, thus reinforcing our conclusions.

Finally, in panel data, errors are often correlated within firms over time, which can bias conventional inference. To address this issue, we used firm-level clustered robust standard errors, which largely correct for residual correlation, although some correlations may persist in extreme cases.

A significant avenue for future research and robustness checking addresses a key limitation of the standard Fixed Effects model: its inability to analyze the direct impact of time-invariant characteristics. While the FE model absorbs these static effects, it is possible to analyze whether these permanent characteristics influence the temporal trends. A relevant methodological extension would be to introduce interaction terms between the time dummy ($\mathbf{D}_{2007}$) and the categorical time-invariant variables such as firm sector or the Firm Age Group (Young, Mature, Old). This technique, which tests for heterogeneity in trends, would reveal whether the single temporal variation in our panel (the 2007 pre-crisis shock) had a significantly different impact on market capitalization based on a structural characteristic. For example, a significant interaction between the dummy variable for the 'Old' age group and $\mathbf{D}_{2007}$ would indicate that firms categorized as Old experienced a more (or less) pronounced change in their market valuation compared to the reference age group (e.g., Young firms) between 2006 and 2007. This extension would provide deeper insight into how the characteristics of permanent firms modulate the effects of dynamic financial and economic changes.

3. Conclusion

In conclusion, our analysis shows that only intra-firm variations in investment and profitability constitute significant determinants of market capitalization over the period studied, whereas pooled OLS models tended to overestimate the influence of structural and governance variables. Importantly, the heterogeneous-effects analysis confirms that profit does not affect all firms in the same way: it is valued more strongly for mature and old firms than for young ones, and even negatively in certain sectors, indicating that investor interpretation depends on both firm maturity and industry-specific dynamics. Together, these results underline that market capitalization is driven not only by dynamic financial mechanisms but also by structural characteristics that shape how these mechanisms operate.The limitations of fixed effects models should be acknowledged, particularly for quasi-invariant variables and time-varying unobservables, yet these models remain the most reliable specification to identify causal relationships within firms and isolate the true economic effects on valuation. These results also point to several avenues for future research. It would be valuable to study the dynamic impact of quasi-invariant structural variables, such as firm age or workforce size, by using longer panels or methods capable of capturing differential effects over time. Moreover, evaluating the effects of more specific macroeconomic shocks or sectoral policies on market capitalization could complement this analysis and offer a more comprehensive view of the determinants of firm value.

Despite the rigor of the panel data analysis, our estimates remain constrained by the ex-post nature of the available variables (realized profit, investment made). In reality, the expectations (or anticipations) of various financial market actors play a decisive role in determining Market Capitalization. A firm's value is fundamentally linked to the perception of its future cash flows, a perception that is constantly adjusted by investors' beliefs regarding the macroeconomic and monetary environment. It would therefore be judicious to direct future research toward the integration of these subjective factors. To achieve this, regular surveys targeting fund managers, analysts, and market participants, specifically concerning central bank policies, could be conducted and their results added to the dataset. The impact of these expectations is clearly evident in the current context where, for instance, the anticipation of a potential US Federal Reserve (Fed) rate cut has recently supported the rebound of certain stock market indices, even following volatility shocks related to political announcements. The inclusion of such sentiment variables would allow researchers to capture short-term market movements that cannot be explained solely by structural fundamentals.

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Appendix

1 : Breusch-Pagan Test

The Breusch-Pagan test was used as a preliminary check for heteroskedasticity, i.e., to assess whether the variance of the residuals is constant across units in a pooled OLS setting. Formally, the hypotheses are:

$$H_0 : \sigma_i^2 = \sigma^2 \quad \text{(homoskedastic residuals)}$$
$$H_1 : \sigma_i^2 \neq \sigma^2 \quad \text{(heteroskedastic residuals)}$$

In our estimation, the test yields a studentized Breusch-Pagan statistic of $BP = 1027.7$ with $df = 18$ and a p-value $< 2.2 \times 10^{-16}$. This extremely small p-value strongly rejects the null hypothesis, indicating that the residual variance is not constant across firms and that heteroskedasticity is present in the pooled OLS model.

2 : Hausman Test

The Hausman test evaluates whether the random-effects (RE) estimator is consistent by examining the correlation between unobserved individual effects and the explanatory variables. The hypotheses are formally stated as follows:

$$H_0 : \text{Cov}(X_{it}, \alpha_i) = 0 \quad \text{(RE estimator is consistent)}$$

$$H_1 : \text{Cov}(X_{it}, \alpha_i) \neq 0 \quad \text{(RE estimator is inconsistent)}$$

In our estimation, the test reports a chi-square statistic of $\chi^2 = 38.723$ with $df = 8$, and a p-value of 5.532×10^{-6}, which is far below conventional significance thresholds. This leads to a clear rejection of the null hypothesis, indicating that the fixed-effects (FE) and random-effects (RE) estimators differ significantly. Consequently, the RE estimator is deemed inconsistent in our setting, and the fixed-effects specification is therefore the appropriate modelling choice for analysing the determinants of wages in our dataset.

It is important to note that, even if the p-value of the Hausman test had technically suggested the consistency of the RE estimator, standard econometric practice for micro-panels like ours generally favors the fixed-effects approach. This preference stems from the inherent difficulty of satisfying the strict exogeneity assumption required for the RE model, which often leads to choosing the FE model a priori to ensure the most robust estimates possible.

3 : Calculation of the Turning Point for Age

To determine the turning point of age, we first take the derivative of the function with respect to age and set it equal to zero:

$$\frac{d(\log_mktcap_i)}{d(\text{age}_i)} = \beta_{\text{age}} + 2\beta_{\text{age}^2} \cdot \text{age}_i = 0$$

The turning point is then calculated as:

$$\text{age}_i^* = -\frac{\beta_{\text{age}}}{2\beta_{\text{age}^2}}$$

With the estimated values:

$$\beta_{\text{age}} = 0.00061 \quad \text{and} \quad \beta_{\text{age}^2} = -0.00001,$$

we obtain:

$$\text{age}_i^* = -\frac{0.00061}{2 \times (-0.00001)} = \frac{0.00061}{0.00002} \approx 30.5$$

Thus, $\log_mktcap_i$ increases with the age of the firm up to approximately 31 years, then starts to slightly decline, reflecting a life-cycle logic: rapid initial growth followed by a phase of maturity and slowdown.

4 : Main Results

Table 2: Estimation of Pooled OLS and Fixed Effects Models

	Modèles d'Estimation					
	Model 1	Model 2	Model 3	Model 4	Within FE	FE + Time
(Intercept)	3.72786***	-4.19259***	-4.19112***	-4.21005***		
	(0.61455)	(0.10336)	(0.10337)	(0.10535)		
profit	0.00364	0.00171***	0.00172***	0.00190***	0.00188***	0.00188***
	(0.00495)	(0.00052)	(0.00052)	(0.00055)	(0.00025)	(0.00025)
log_inv	0.00774	0.02104***	0.02108***	0.03618***	0.01203***	0.01204***
	(0.01244)	(0.00107)	(0.00107)	(0.00269)	(0.00135)	(0.00135)
log_shortdebt	0.29524	-0.00390	-0.00389	-0.01203	-0.00054	-0.00076
	(0.19925)	(0.02154)	(0.02153)	(0.02229)	(0.00228)	(0.00231)
log_emp		0.91211***	0.91208***	0.91188***		
		(0.00350)	(0.00350)	(0.00350)		
log_nboard		0.02576	0.02606	0.02374	-0.00301	-0.00302
		(0.01656)	(0.01655)	(0.01645)	(0.00195)	(0.00195)
wboard_ratio		0.00780	0.00797	0.01080	0.00375	0.00370
		(0.00636)	(0.00635)	(0.00680)	(0.01641)	(0.01641)
highdegree_emp_ratio			-0.03533*	-0.03541*	0.02136	0.02159
			(0.01959)	(0.01947)	(0.01926)	(0.01933)
age				0.00061**		
				(0.00031)		
age_squared				-0.00001*	0.00000	0.00000
				(0.00000)	(0.00000)	(0.00001)
factor(sector)2				0.00547		
				(0.00507)		
factor(sector)3				-0.00305		
				(0.00345)		
factor(sector)4				0.00017		
				(0.00393)		
factor(sector)5				0.00039		
				(0.00403)		
factor(sector)6				-0.00295		
				(0.00410)		
factor(sector)7				-0.00732**		
				(0.00353)		
factor(sector)8				0.04449***		
				(0.00717)		
factor(sector)9				-0.00110		
				(0.00427)		
factor(sector)10				-0.00522		
				(0.00393)		
factor(year)2007						0.00015
						(0.00037)
Observations	7,977	7,977	7,977	7,977	7,977	7,977
R^2	0.000	0.992	0.992	0.992	0.067	0.067
R^2 Adj.	-0.000	0.992	0.992	0.992	-0.875	-0.875
Log.Lik.	-10193.292	9126.560	9128.126	9170.119		

Note: *p<0.1; **p<0.05; ***p<0.01

5 : Heterogeneous Effects and Robustness

Table 3: Comparison of 4 Fixed Effects Models with Clustered Robust SE

	log_mktcap			
	Model 1	Model 2	Model 3	Model 4
	(1)	(2)	(3)	(4)
profit	0.000141	0.000141	0.000451	0.000446
	(0.000581)	(0.000581)	(0.000687)	(0.000683)
age_mature	-0.001771**	-0.001798**	-0.001634*	-0.001540*
	(0.000855)	(0.000898)	(0.000856)	(0.000863)
age_old	-0.001157	-0.001198	-0.000940	-0.000834
	(0.001130)	(0.001201)	(0.001170)	(0.001182)
log_inv	0.012078***	0.012079***	0.012086***	0.012053***
	(0.001353)	(0.001352)	(0.001358)	(0.001366)
log_shortdebt	-0.000635	-0.000689	-0.000952	-0.001050
	(0.002274)	(0.002315)	(0.002312)	(0.002326)
log_nboard	-0.003158	-0.003159	-0.003156	-0.003148
	(0.001953)	(0.001951)	(0.001950)	(0.001964)
wboard_ratio	0.004034	0.004030	0.003789	0.290975
	(0.016383)	(0.016384)	(0.016262)	(0.677158)
highdegree_emp_ratio	0.021542	0.021588	0.024061	0.024081
	(0.019289)	(0.019334)	(0.019403)	(0.019387)
age_squared	0.000004	0.000003	0.000004	0.000004
	(0.000003)	(0.000006)	(0.000006)	(0.000006)
factor(year)2007		0.000037	0.000007	0.000016
		(0.000384)	(0.000380)	(0.000380)
profit:age_mature	0.001442***	0.001441***	0.001417***	0.001432***
	(0.000434)	(0.000434)	(0.000436)	(0.000436)
profit:age_old	0.001779***	0.001778***	0.001793***	0.001799***
	(0.000541)	(0.000542)	(0.000536)	(0.000531)
profit:factor(sector)2			-0.000864	-0.000871
			(0.000752)	(0.000752)
profit:factor(sector)3			-0.000750	-0.000740
			(0.000720)	(0.000724)
profit:factor(sector)4			0.000064	0.000059
			(0.000845)	(0.000844)
profit:factor(sector)5			-0.000905	-0.000896
			(0.000714)	(0.000709)
profit:factor(sector)6			0.000138	0.000147
			(0.001169)	(0.001157)
profit:factor(sector)7			0.001563*	0.001564*
			(0.000934)	(0.000930)
profit:factor(sector)8			0.000578	0.000612
			(0.001371)	(0.001362)
profit:factor(sector)9			-0.001638***	-0.001640***
			(0.000587)	(0.000591)
profit:factor(sector)10			-0.001115	-0.001113
			(0.000961)	(0.000967)
log_emp:wboard_ratio				-0.030081
				(0.069401)
Observations	7,977	7,977	7,977	7,977
R^2	0.067694	0.067695	0.071113	0.071667

Note: *p<0.1; **p<0.05; ***p<0.01

6 : R-Code